

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

Solutions:

QUESTION 1: Debugging the Network Configuration

Given:

- IP Address = 192.168.10.65
- Subnet Mask = 255.255.255.192 (/26)
- Default Gateway = 192.168.10.1

Step 1: Identify the Subnet

Subnet Mask = /26

Subnet Increment= 64

Subnet Ranges

192.168.10.0 – 192.168.10.63

192.168.10.64 – 192.168.10.127

192.168.10.128 – 192.168.10.191

192.168.10.192 – 192.168.10.255

Configured IP Address

192.168.10.65

Therefore,

The IP Address belongs to

192.168.10.64/26

Step 2: Determine Network Information

Network Address

192.168.10.64

Broadcast Address

192.168.10.127

Valid Host Range

192.168.10.65 – 192.168.10.126

Step 3: Verify IP Address and Gateway

Configured IP Address

192.168.10.65

Status

✓ Valid Host Address

Configured Gateway

192.168.10.1

Gateway belongs to

192.168.10.0/26

Current Workstation belongs to

192.168.10.64/26

Therefore,

Gateway and workstation are in different subnets.

Step 4: Identify the Configuration Error

Error Identified

- IP Address belongs to subnet
192.168.10.64/26
- Default Gateway belongs to subnet
192.168.10.0/26

Because both devices are in different subnets,
the workstation cannot communicate with the gateway.

Step 5: Optimize the Configuration

Correct Configuration

IP Address
192.168.10.66

Subnet Mask
255.255.255.192 (/26)

Default Gateway
192.168.10.65

(Any valid router interface within
192.168.10.64/26 can be used.
Here 192.168.10.65 is selected.)

Step 6: Calculate Usable Hosts

Host Bits
 $32 - 26 = 6$

Usable Hosts
 $2^6 - 2 = 64 - 2$
= 62 Hosts

Final Result Table

Description	Result
Original IP Address	192.168.10.65
Original Default Gateway	192.168.10.1
Subnet	192.168.10.64/26
Network Address	192.168.10.64
Broadcast Address	192.168.10.127
Valid Host Range	192.168.10.65 – 192.168.10.126
Is IP Address Valid?	Yes
Is Gateway Valid?	No
Configuration Error	Default Gateway belongs to a different subnet
Optimized IP Address	192.168.10.66
Optimized Default Gateway	192.168.10.65
Usable Host Addresses	62

Interpretation of Results

The workstation IP address 192.168.10.65 belongs to the subnet 192.168.10.64/26, while the configured gateway 192.168.10.1 belongs to a different subnet (192.168.10.0/26). Because the IP address and default gateway are not in the same subnet, communication fails.

After correcting the gateway to an address within the same subnet (for example, 192.168.10.65) and assigning a valid host IP such as 192.168.10.66, the workstation can communicate successfully with other devices. The corrected subnet provides 62 usable host addresses, ensuring efficient and reliable network communication.

QUESTION 2: Debugging IPv6 Configuration

Given:

- Device A = 2001:db8::ff00:42:8329/64
- Device B = 2001:db8:0:0:1::1/64

Step 1: Expand Both IPv6 Addresses

Device A

Compressed Address

2001:db8::ff00:42:8329

Expanded Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

Compressed Address

2001:db8:0:0:1::1

Expanded Address

2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Determine the Network Prefix

Prefix Length

/64

Device A

Network Prefix

2001:db8:0000:0000::/64

Device B

Network Prefix

2001:db8:0000:0000::/64

Step 3: Verify Whether Both Devices Belong to the Same Subnet

Device A Prefix

2001:db8:0000:0000::/64

Device B Prefix

2001:db8:0000:0000::/64

Result

✓ Both devices belong to the same IPv6 subnet.

Step 4: Identify the Configuration Error

Both devices have the same /64 network prefix.

Therefore,

The IPv6 addressing and prefix are correct.

The communication failure is NOT caused by IPv6 addressing.

Possible causes include:

- Incorrect default gateway
- Router configuration error
- Firewall blocking traffic
- Neighbor Discovery Protocol (NDP) failure
- Interface configuration problem

Step 5: Optimize the IPv6 Configuration

Recommended Configuration

Device A
2001:db8:0:0::10/64
Device B
2001:db8:0:0::20/64

Default Gateway
2001:db8:0:0::1/64

Both devices now belong to the same subnet with unique host addresses.

Step 6: Calculate Available Host Addresses

Host Bits
 $128 - 64 = 64$

Available Host Addresses = 2^{64}
= 18,446,744,073,709,551,616

Final Result Table

Description	Result
Device A (Expanded)	2001:0db8:0000:0000:0000:ff00:0042:8329
Device B (Expanded)	2001:0db8:0000:0000:0001:0000:0000:0001
Device A Network Prefix	2001:db8:0000:0000::/64
Device B Network Prefix	2001:db8:0000:0000::/64
Do Both Devices Belong to the Same Subnet?	Yes
Addressing Error	None
Possible Cause of Communication Failure	Gateway, Router, Firewall, or NDP issue
Optimized Device A Address	2001:db8:0:0::10/64
Optimized Device B Address	2001:db8:0:0::20/64
Optimized Default Gateway	2001:db8:0:0::1/64
Available Host Addresses	18,446,744,073,709,551,616 (2^{64})

Interpretation of Results

Both IPv6 devices share the same /64 network prefix (2001:db8:0000:0000::/64), which means they are in the same subnet. Therefore, the communication failure is not due to an IPv6 addressing or prefix mismatch. The issue is likely related to router configuration, default gateway settings, firewall rules, or Neighbor Discovery Protocol (NDP). Assigning unique IPv6 addresses within the same subnet and configuring the correct default gateway ensures reliable communication. A /64 IPv6 subnet supports 2^{64} (18,446,744,073,709,551,616) possible host addresses, making it suitable for large-scale IoT and enterprise networks.

QUESTION 4: Debugging IPv6 Configuration

- Given:
- Device A = 2001:db8::ff00:42:8329/64
 - Device B = 2001:db8:0:0:1::1/64

Step 1: Expand Both IPv6 Addresses

Device A

Compressed Address
2001:db8::ff00:42:8329

Expanded Address
2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

Compressed Address
2001:db8:0:0:1::1

Expanded Address
2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Determine the Network Prefix

Prefix Length = /64

Device A Network Prefix
2001:db8:0000:0000::/64

Device B Network Prefix
2001:db8:0000:0000::/64

Step 3: Verify Whether Both Devices Belong to the Same Subnet

Both devices have the same network prefix.

Result:
✓ Both devices belong to the same subnet.

Step 4: Identify the Configuration Error

IPv6 addressing is correct.
Possible reasons for communication failure are:

- Incorrect Default Gateway
- Router Configuration Error
- Firewall Blocking Traffic
- Neighbor Discovery Protocol (NDP) Failure
- Interface Configuration Error

Step 5: Optimize the IPv6 Configuration

Recommended Configuration
Device A
2001:db8:0:0::10/64
Device B
2001:db8:0:0::20/64

Default Gateway
2001:db8:0:0::1/64

Step 6: Calculate Available Host Addresses

Host Bits
= 128 – 64
= 64

Available Host Addresses
= 2^64
= 18,446,744,073,709,551,616

Final Result Table

Description	Result
Device A (Expanded)	2001:0db8:0000:0000:0000:ff00:0042:8329
Device B (Expanded)	2001:0db8:0000:0000:0001:0000:0000:0001

Description	Result
Device A Network Prefix	2001:db8:0000:0000::/64
Device B Network Prefix	2001:db8:0000:0000::/64
Do Both Devices Belong to the Same Subnet?	Yes
Addressing Error	None
Possible Cause of Communication Failure	Gateway, Router, Firewall, or NDP Issue
Optimized Device A Address	2001:db8:0:0::10/64
Optimized Device B Address	2001:db8:0:0::20/64
Optimized Default Gateway	2001:db8:0:0::1/64
Available Host Addresses	18,446,744,073,709,551,616 (2^64)

Interpretation of Results

Both devices belong to the same IPv6 subnet because they share the network prefix 2001:db8:0000:0000::/64. Therefore, the communication failure is not caused by an addressing or prefix mismatch. The issue is likely related to the router configuration, default gateway, firewall settings, or Neighbor Discovery Protocol (NDP). Assigning unique IPv6 addresses within the same subnet and configuring the correct default gateway ensures successful communication. A /64 IPv6 subnet supports 2^64 host addresses, making it suitable for large enterprise and IoT networks.

QUESTION 5: Debugging the Routing Configuration

Given:

Routing Table

Destination Network Next Hop
----- -----
192.168.1.0/24 10.0.0.2
192.168.2.0/24 10.0.0.3
Default Route 10.0.0.1

Destination IP Address

192.168.2.100

Step 1: Identify the Matching Routing Entry

Destination IP Address

192.168.2.100

Subnet Mask

/24

Network Address

192.168.2.0/24

Using the Longest Prefix Match,

Matching Routing Entr

192.168.2.0/24

Step 2: Determine the Next-Hop Router

Routing Table Entry

192.168.2.0/24

Next Hop

10.0.0.3

Therefore,

The router forwards the packet to

10.0.0.3

Step 3: Verify the Destination Subnet

Destination IP
192.168.2.100

Network Address
192.168.2.0/24

Host Range
192.168.2.1 – 192.168.2.254

Broadcast Address
192.168.2.255

Result
✓ The destination IP belongs to the intended subnet.

Step 4: Identify Routing Configuration Errors

The routing table contains a correct entry for
192.168.2.0/24

Therefore,
No routing table error exists.
If packets are not reaching the destination,
possible causes include:

- Next-hop router (10.0.0.3) is unreachable.
- Interface connected to 10.0.0.3 is down.
- Incorrect routing configuration on the next-hop router.
- Firewall or ACL blocking traffic.
- Destination host is offline.

Step 5: Optimize the Routing Table

Recommended Optimization

- ✓ Verify connectivity to 10.0.0.3.
- ✓ Ensure the interface is active.
- ✓ Configure route summarization if multiple routes exist.
- ✓ Enable dynamic routing protocols such as OSPF or EIGRP for automatic route updates.
- ✓ Regularly monitor routing table entries.

Step 6: Identify the Default Route

If no routing table entry matches the destination network,
the router forwards the packet using

Default Route
0.0.0.0/0

Next Hop
10.0.0.1

Final Result Table

Description	Result
Destination IP Address	192.168.2.100
Matching Routing Entry	192.168.2.0/24
Longest Prefix Match	192.168.2.0/24
Selected Next-Hop Router	10.0.0.3

Description	Result
Destination Subnet	192.168.2.0/24
Destination Host Valid?	Yes
Routing Table Error	None
Possible Cause of Failure	Next-hop unreachable, Interface down, ACL/Firewall blocking traffic, or Destination host unavailable
Optimized Routing	Verify next-hop, enable route summarization, use dynamic routing protocols (OSPF/EIGRP)
Default Route	0.0.0.0/0 via 10.0.0.1

Interpretation of Results

The destination IP address 192.168.2.100 belongs to the subnet 192.168.2.0/24. Using the longest-prefix match, the router correctly selects the routing entry 192.168.2.0/24 and forwards the packet to the next-hop router 10.0.0.3.

Since the routing table is correct, the communication failure is likely caused by issues such as an unreachable next-hop router, interface failure, firewall rules, or problems on the destination host. If no matching destination exists, the router forwards the packet through the default route (0.0.0.0/0) using the next-hop router 10.0.0.1. Proper routing optimization improves packet forwarding efficiency and network reliability

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

